Physics 4730, Fall 2026 Homework #4

Due Sunday November 1st

1. Write a Python script **quantile.py** which reads in an unknown random variable from a one-column file **rvvec.dat** and estimates the α quantile of the data and it's 95% confidence interval using the empirical likelihood method. α should be passed to your script on the command line.

   For example, if I call your script as follows:

   ```
   belz@canis > python3 quantile.py 0.5
   ```

   then the script should read **rvvec.dat** (assumed to be in the same directory as your script), compute the median (50% of the data below this value of the RV) and provide a range of the RV in which we can be 95% certain the "true" median lies.

   ```
   The 0.500 quantile of the data is: x.xxxxxx
   The 95 pct CI is (y.yyyyyy, z.zzzzzz)
   ```

   It is likely useful, but not required, that you produce a plot to confirm your calculation is doing something sensible.

2. In lecture and lab we considered the *Kolmogorov-Smirnov* (KS) test, for comparison of a single dataset with a completely-specified theoretical model, or for comparison of two datasets. This test was limited to one-dimensional RVs, as the CDF is not well-defined in more than one dimension. For this assignment, following Press and Teukolsky[1], we consider a modification to the KS procedure to compare two-dimensional RVs.

   Your task is to write a Python function in **2dks.py** to compute p-value for two two-dimensional Gaussian RVs. Each RV should have 100 or so data points. You are then to use this function to evaluate how average p-value changes as the relative *widths* of two 2D Gaussians (with the same mean) are varied. Specifically, set $\sigma_{x1} = \sigma_{y1} = 1$ for one 2D RV, then vary $\sigma_{x2} = \sigma_{y2}$ for the second RV. On both the low- and high-ends, for what values of $\sigma_{x2} = \sigma_{y2}$ will the average p-value be below 5%?

   The procedure for performing this calculation is contained in Press and Teukolsky, including some code (albeit in FORTRAN). I have also provided the data used in the example given in the paper in **ptdata.squares.dat** and **ptdata.triangles.dat**: It may be useful in developing your code to have data for which the "answer" is known. (Though the instructor will offer some caveats about this in class.)

Submit your assignment by 11:59 PM on Sunday, November 1:

submit p4730 hw04 quantile.py 2dks.py

[1] W.H. Press and S.A. Teukolsky, *K-S Test for Two-Dimensional Data*, Computers in Physics **2**, 74 (1988).